

On the efficiency of mandatory retirement savings under endogenous borrowing constraints

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Abstract

Mandatory contributions to retirement savings accounts may tighten existing borrowing constraints, forcing individuals to forgo profitable investment options. This welfare-detrimental effect can be offset if retirement savings are allowed to serve as collateral. Moreover, some credit market imperfections may disappear altogether if the latter policy is combined with a pension system with unconditional basic savings.

KEYWORDS: pensions, mandatory savings, borrowing constraints, collateral. JEL CODES: D14, E21, G11, H55

1 Introduction

Mandatory contributions to individual retirement savings accounts are commonly justified as a commitment device to postpone consumption for old age.¹ However, this device will only be effective as long as some borrowing constraint prevents individuals from undoing the increase in savings. Nonetheless, the literature assumes that the borrowing constraint is exogenous.² In this paper, I endogenize the borrowing constraints and show that mandatory savings may exacerbate the existing inefficiencies of the credit market. The mechanism is that mandatory contributions restrict the liquidity of non-retired individuals who need to borrow in order to invest.³ This negative effect can be offset if the retirement savings can serve as collateral.⁴ Furthermore, if the policy of allowing retirement savings to serve as collateral is combined with a policy of transferring the contributions of high earners to the retirement accounts of low earners, then the inefficiencies of the credit market can be overcome.

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¹This is due to myopia or time-inconsistent preferences. See Feldstein (1985), Laibson et al. (1998) or Diamond and Köszegi (2003) for some references.

²See Feldstein and Liebman (2002), İmrohoroglu et al. (2003), Cremer et al. (2008) and Dybvig and Liu (2010) for examples of life cycle models where borrowing constraints are exogenous.

³As shown by Coeurdacier et al. (2015), individuals tend to be net borrowers before reaching their middle age. This observation led Andersen and Bhattacharya (2017) to correctly ask why young borrowers are forced to save for their retirement. My answer to this question contrasts with theirs. Recent work on withdrawals and loans from retirement accounts also highlights the liquidity role of these otherwise illiquid savings; see Derby et al. (2022) and Garivaltis (2021).

⁴To my knowledge, no previous analysis of mandatory contributions to individual retirement accounts acknowledges the role of collateral in borrowing contracts. For an analysis of this type of contracts, see Geanakoplos (2010) and Simsek (2013).

2 The Model

Consider an economy in discrete time with three generations in each period: young (y), middle aged (m) and old (o). Each generation is composed of a continuum of risk-neutral individuals with measure one. Their instantaneous utility depends exclusively on consumption. If c_i denotes consumption at age $i \in \{y, m, o\}$ and their discount factor is β , then their payoff function is given by the following function:

$$c_y + \beta c_m + \beta^2 c_o$$

Individuals work while they are not old, supplying labor inelastically. The probability for a young individual of having labor earnings below w_y is given by $G(w_y)$, where the support of G is $[0, w_+]$, where w_+ is positive, and where G is strictly increasing on its support. Conditional on a realization w_y , the probability of earning w_m or less at middle age is $H(w_m|w_y)$. Individuals at age $i \in \{y, m\}$ are obliged to save a fraction $\tau_i \in [0, 1]$ of their labor earnings into individual retirement accounts. These savings are only available for consumption when individuals grow old.

Let R_f denote the gross return on the safe asset. Both individuals and retirement funds are able to save by investing in this safe asset or by lending to other individuals. Individuals with a collateral coverage ratio v —the share of the loan’s value that is backed by pledged assets—can pledge to pay $R(v)$ on each dollar borrowed. Nonetheless, they are subject to limited liability. Creditors cannot seize any wages in case of default, but can seize an exogenous fraction $\eta \in [0, 1]$ of the retirement savings.⁵ In equilibrium, creditors will have to be indifferent between investing at a return R_f and lending at a rate $R(v)$. Still, credit might be rationed for moral hazard reasons explained further below. Furthermore, in equilibrium the consumer has to be indifferent between the allocation of consumption across different periods. Therefore, $R_f = \beta^{-1}$.

Young individuals may become net borrowers in order to invest in an option only available to them (think of it as education). This investment yields a stochastic payoff z , which is realized when individuals reach middle age. The support of z is given by $\{0, \pi\}$, where $\pi > 0$. The distribution of z depends on the investment done at young age, which is denoted by x . A chance of earning π requires an investment of at least $F > 0$. Still, there is a chance $\epsilon \in (0, 1)$ of failing to achieve a payoff π . The probability of a payoff π at middle age is given by:

$$P(z = \pi \mid x) = \begin{cases} 0 & \text{if } x < F, \\ 1 - \epsilon & \text{if } x \geq F \end{cases} \quad (1)$$

Hence, the expected gross return on this alternative investment is given by:

$$E\left(\frac{z}{x}\right) = \begin{cases} 0 & \text{if } x < F, \\ \frac{(1-\epsilon)\pi}{x} & \text{if } x \geq F \end{cases} \quad (2)$$

The following assumption has two roles. The first inequality guarantees that the alternative investment has positive net present value, so all young agents would like to undertake it if they can finance it. The second inequality guarantees that this net present value is not so large as to eliminate borrowing constraints altogether:

Assumption 1. $F < \beta(1 - \epsilon)\pi < 2F$

Consider an individual for whom $(1 - \tau_y)w_y < F$. In order to invest in the alternative asset, she would have to borrow the following amount:

$$b(w_y) = F - (1 - \tau_y)w_y. \quad (3)$$

⁵See Geanakoplos (2010) and Simsek (2013) for an analysis of collateralized loans.

Let $B(w_y) := \tau_y w_y$ denote the balance in her retirement account. Her collateral coverage ratio is given by:

$$v(w_y) = \frac{\eta B(w_y)}{b(w_y)} \quad (4)$$

When no confusion arises, v denotes a generic collateral coverage ratio, while $v(w_y)$ denotes the coverage ratio of an individual with young earnings w_y . If she is able to borrow $b(w_y)$ and she invests F in the alternative asset, her middle-age consumption will depend on the realization z for her investment payoff and the realization w_m for her middle-age labor earnings. Let s_m be her voluntary savings at middle age. Use $(\cdot)^+$ to denote the maximum between zero and the argument inside the brackets. Since creditors cannot seize labor earnings, her budget constraint at middle age is given by:

$$c_m + s_m = (1 - \tau_m)w_m + (z - R(v(w_y))b(w_y))^+. \quad (5)$$

If z turns out to be zero, the individual defaults on her debt. Since young retirement savings have grown to $R_f B(w_y)$ by middle age, creditors may seize up to $R_f \eta B(w_y)$ from the pledgeable part of her retirement savings. The remaining balance then grows until old age, so her consumption at old age is given by:

$$c_o = \begin{cases} R_f (s_m + \tau_m w_m + R_f B(w_y)) & \text{if } z = \pi, \\ R_f \left(s_m + \tau_m w_m + (1 - \eta)R_f B(w_y) + (R_f \eta B(w_y) - R(v(w_y))b(w_y))^+ \right) & \text{if } z = 0. \end{cases} \quad (6)$$

Since the alternative investment is no longer available when she reaches middle age, the best gross return she can get on both her mandatory retirement savings and her voluntary savings is R_f . Given her risk-neutrality and her discount factor, she is indifferent regarding her voluntary savings, as shown by adding the discounted consumption in (6) to the consumption in (5):

$$c_m + \beta c_o = w_m + (1 - \eta)R_f B(w_y) + (\eta R_f B(w_y) + z - R(v(w_y))b(w_y))^+. \quad (7)$$

Furthermore, her contribution τ_m at middle age is inconsequential for her welfare.

In equilibrium, creditors have to be indifferent regarding lending to a young individual with a collateral coverage v and investing in the asset that yields R_f . To see this, consider one dollar lent to a young individual. If the investment succeeds, which occurs with probability $1 - \epsilon$, the creditor receives the promised repayment $R(v)$. If the investment fails, which occurs with probability ϵ , the individual defaults and the creditor can recover at most the middle-age value of pledgeable collateral per dollar borrowed, $R_f v$. Since the debt contract never pays more than the promised repayment, the creditor's recovery in default is $\min\{R_f v, R(v)\}$. Risk-neutral creditors therefore break even when their expected repayment equals the risk-free payoff:

$$R_f = (1 - \epsilon)R(v) + \epsilon \min\{R_f v, R(v)\}, \quad (8)$$

Therefore, the lending rate must have a risk premium over the risk-free rate:

$$R(v) = \begin{cases} \left(\frac{1 - \epsilon v}{1 - \epsilon} \right) R_f & \text{if } v < 1, \\ R_f & \text{otherwise.} \end{cases} \quad (9)$$

Let $V(w_y)$ denote the expected payoff for a young individual endowed with w_y in labor earnings. Use $\mathcal{N} := \beta(1 - \epsilon)\pi - F > 0$ to denote the expected net present value of the alternative investment. Equations (3), (4), (7) and (9) imply that, *conditional on investing in the alternative asset*, her

lifetime payoff is the expected present value of her wages plus the net present value of the alternative investment. The zero-profit condition for lenders implies that the present value of expected repayments equals the amount borrowed, so the borrower captures the project's net present value:

$$V(w_y) = w_y + \beta E(w_m | w_y) + \mathcal{N} \quad (10)$$

Nonetheless, it is assumed that x is not observable, so creditors cannot force any borrower to invest in the alternative asset. This creates a moral hazard problem since the borrower can consume $b(w_y)$ in her youth and face a penalty limited to $\eta B(w_y)$. Therefore, creditors will only lend to individuals who satisfy the following incentive-compatibility constraint:

$$V(w_y) \geq w_y + \beta E(w_m | w_y) + b(w_y) - \eta B(w_y) \quad (11)$$

Definition 1. *Given policy parameters (τ_y, τ_m, η) , an equilibrium consists of borrowing, investment, consumption, and saving choices for individuals and a lending rate schedule $R(v)$ such that: individuals maximize expected lifetime payoff subject to the budget constraints (5)–(6) and limited liability; borrowers repay when $z = \pi$ and default when $z = 0$; lenders break even against the safe asset for every collateral coverage ratio v at which they lend, so that (8) holds; and young individuals invest in the alternative asset whenever they can finance F and satisfy the incentive-compatibility constraint (11).*

3 Results

Substituting (10) into (11) implies:

$$b(w_y) \leq \mathcal{N} + \eta B(w_y). \quad (12)$$

In other words, investing in the alternative asset requires a downpayment equal to $F - \mathcal{N} - \eta B(w_y)$. However, disposable income for young individuals is only $(1 - \tau_y)w_y$. Using $B(w_y) = \tau_y w_y$, no individual with earnings below a certain threshold will be able to borrow enough to invest in the alternative asset. This threshold is given by:

$$\underline{w}_y := \frac{F - \mathcal{N}}{1 - (1 - \eta)\tau_y} \quad (13)$$

The upper bound in Assumption 1 implies $F > \mathcal{N}$. Thus, if the support of G includes 0, the equilibrium is inefficient.

Since $\mathcal{N} > 0$, aggregate welfare is decreasing in the fraction of young individuals not able to get enough credit:

$$\int V(w_y) G(dw_y) = E(w_y) + \beta E(w_m) + (1 - G(\underline{w}_y)) \mathcal{N} \quad (14)$$

Proposition 1. *Aggregate welfare is weakly decreasing in $(1 - \eta)\tau_y$. It is strictly decreasing whenever the implied threshold $\underline{w}_y$ lies in the support of G .*

Proof. By (13) and $F > \mathcal{N}$, $\underline{w}_y$ increases with $(1 - \eta)\tau_y$. Since G is weakly increasing, the fraction of young individuals below the threshold weakly increases. Equation (14) and $\mathcal{N} > 0$ then imply that aggregate welfare weakly decreases. Since G is strictly increasing on its support, the decrease is strict whenever $\underline{w}_y$ lies in the support of G . $\square$

Notice how any increase in the defined contribution τ_y works: On one hand, it decreases disposable income, forcing young individuals to demand more borrowing. On the other, a fraction η of τ_y can be pledged as collateral, decreasing the downpayment. Hence the only policy parameter

relevant for welfare is the fraction of mandatory contributions $(1 - \eta)\tau_y$ that cannot serve as collateral. If $\eta = 1$ and all retirement savings can serve as collateral, then all mandatory contributions are irrelevant. Nonetheless, as long as $\eta < 1$, an increase in the mandatory contribution τ_y tightens credit, limits investment and decreases welfare. This welfare-detrimental effect is increasing in the share $(1 - \eta)$ of mandatory savings not allowed to serve as collateral.

4 Unconditional basic savings

The model also suggests a way in which mandatory retirement savings can *increase* efficiency. Equation (13) shows that, as long as $F - \mathcal{N}$ is positive, some low-wage young individuals cannot finance the alternative investment. These individuals would like to borrow more than creditors are willing to lend them, even when there are no mandatory contributions or when all retirement savings can serve as collateral.

The source of the inefficiency is not mandatory saving per se, but the lack of pledgeable resources among low-wage young individuals. Equation (12) shows that the minimum collateral balance needed to remove this constraint for every young individual is $F - \mathcal{N}$. A natural response is therefore to guarantee a minimum retirement balance of exactly this amount that can be used as collateral.

Consider an *unconditional basic savings* system with the following features:

1. A tax rate ϕ is levied on young and middle-age labor earnings above $F - \mathcal{N}$. The tax rate is set so that tax revenue funds an unconditional retirement saving of $F - \mathcal{N}$ for each young individual:

$$\phi = \frac{\int_0^\infty (F - \mathcal{N}) dG(w_y)}{\int_{F-\mathcal{N}}^\infty w_y dG(w_y) + \int_0^\infty \int_{F-\mathcal{N}}^\infty w_m dH(w_m|w_y) dG(w_y)} \quad (15)$$

The denominator in (15) is the corresponding tax base, including young earnings above $F - \mathcal{N}$ and middle-age earnings above $F - \mathcal{N}$.

2. Young individuals make additional mandatory contributions only on earnings above $F - \mathcal{N}$:

$$T(w_y) = \tau_y (w_y - (F - \mathcal{N}))^+. \quad (16)$$

3. The retirement balance of a young individual is therefore:

$$B(w_y) = F - \mathcal{N} + \tau_y (w_y - (F - \mathcal{N}))^+ \quad (17)$$

If at least $F - \mathcal{N}$ of the retirement balance can be pledged as collateral, the incentive-compatibility constraint no longer prevents any young individual from financing the alternative investment. All young individuals can then invest in the alternative asset.⁶

A natural concern regarding the unconditional basic savings system is whether there are enough resources to fund such a system. For example, Equation (15) implies that, for ϕ to be less than one, labor earnings among individuals above $F - \mathcal{N}$ have to be sufficiently high. Given the constraints on obtaining funds from labor income, a contribution from capital income should not be ruled out. Especially in the context of increased automation, funding unconditional basic savings through taxes on capital earnings may dampen the increase in wealth inequality, although the overall effect of such a funding scheme on savings and investment might be ambiguous.

⁶The insight regarding the effect of wealth distribution on the proportion of credit-constrained individuals is due to Galor and Zeira (1993) and Banerjee and Newman (1993). See also Ghatak and Jiang (2002) for a simple model.

5 Final remarks

Mandatory savings change the liquidity and collateral position of borrowers, narrowing their investment opportunities. When retirement balances cannot be pledged, higher contributions tighten borrowing constraints and reduce investment in positive-net-present-value opportunities. When retirement balances can serve as collateral, this distortion is mitigated.

This provides an argument in favor of allowing retirement savings to serve as collateral for loans that finance long-lived investments such as education or housing. Such a policy differs from an early withdrawal because the retirement balance remains in the account unless the borrower defaults. Real-world pension systems already allow related forms of restricted access: Singapore’s Central Provident Fund allows Ordinary Account savings to be used for housing purchases and mortgage payments, while qualified retirement plans in the United States may provide participant loans, with longer repayment horizons for loans used to purchase a principal residence.⁷ The policy implication is therefore not to make retirement savings fully liquid, but to allow pledgeability when the loan finances assets that relax life-cycle investment constraints.

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⁷See CPF Board, “Using your CPF to buy a home,” <https://www.cpf.gov.sg/member/home-ownership/using-your-cpf-to-buy-a-home>, and Internal Revenue Service, “Retirement plans FAQs regarding loans,” <https://www.irs.gov/retirement-plans/retirement-plans-faqs-regarding-loans>.

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